

May-26-0025

CSPC-613 (Compiler Design)

B.Tech-6th (NEP) CSE, AI&ML (Session 2022-26) / B.Tech-6th NEP CSE/CSE(AI&DS)/CSE(AI&ML)

Time : 3 Hours

Max. Marks : 60

Note : The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, select one question each from Section A, B, C, D. Section E (Question 9) is compulsory.

SECTION - A

1. (a) Define compiler, interpreter, and highlight their differences with suitable examples. **(6)**
(b) Explain the various phases of compilation with neat diagrams. **(6)**

OR

2. (a) Describe the role of the Lexical Analyzer and explain input buffering techniques. **(6)**
(b) Explain the construction of a DFA from a Regular Expression with a suitable example. **(6)**

SECTION - B

3. (a) What is ambiguity in grammars? Discuss methods to remove ambiguity. **(6)**
(b) Explain the Recursive Descent Parser with a working example. **(6)**

OR

4. (a) Differentiate between an LL(1) parser and an LR(0) parser with examples. **(6)**
(b) Explain the role of the YACC tool in syntax analysis. **(6)**

SECTION - C

5. (a) Explain syntax-directed definitions and evaluation orders. **(6)**
(b) Construct a Three Address Code for the expression: $A = (B + C) * (D - E)$. **(6)**

OR

6. (a) Draw and explain a syntax tree for the statement: `while (a < b) { a = a + 1; }`. **(6)**
(b) Write short notes on type checking and translation of declarations. **(6)**

SECTION - D

7. (a) Explain the need for runtime memory management in detail. **(6)**
(b) Discuss stack allocation and heap management with diagrams. **(6)**

OR

8. (a) What are the principal sources of code optimization? Explain with examples. **(6)**
(b) Explain DAG representation and its role in optimization. **(6)**

SECTION - E

(Compulsory)

9. (a) Define bootstrapping in compiler design.
(b) Differentiate between top-down and bottom-up parsing.
(c) What is peep-hole optimization? Give an example.
(d) Mention any two issues in code generation.

(e) Write the advantages of minimizing a DFA.

(f) Explain various types of peep-hole optimizations. **(6×2=12)**

Transcribed from a photograph of the original answer/question booklet. A few numeric values, table entries and one section could not be fully verified due to image quality (angle/blur/overlap with adjacent papers) — please cross-check against your original copy.